Babel: Reaching Technology Democratization with Translation Layers

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Abstract

Large language models (LLMs) are increasingly presented as universal technologies capable of transforming communication, education, health care, and economic productivity for all. However, a fallacy of LLMs is that their benefits remain distributed unevenly across linguistic communities. While these systems have indeed performed strongly in English, they frequently underperform in non-English and low-resource languages, producing unnatural phrasing, grammatical errors, hallucinations, cultural misalignment, and weaker safety behavior. This paper argues that such disparities should be understood not merely as technical limitations, but as a failure of technology democratization. When AI systems function reliably for English speakers while offering degraded or unsafe outputs for others, they reproduce global inequalities in access to information, labor opportunities, and digital participation. Drawing on recent research and policy discussions, this paper examines the structural causes of multilingual underperformance, including unequal training data, English-centric evaluation practices, translation-based shortcuts, and the cultural values embedded in model development. We present empirical evidence from paired prompt experiments comparing English, Chinese, and Nepali inputs across both image-generation and text-generation tasks—demonstrating that English prompts yield measurably superior outputs across all modalities and evaluation metrics. Critically, M-Prometheus does not natively support Nepali evaluation, a fact we treat not as a methodological inconvenience but as direct evidence of the multilingual gap this paper documents: low-resource language communities are underserved not only by AI systems, but by the tools used to measure those systems.

1 Introduction

The rapid rise of large language models (LLMs) has reshaped public and scholarly conversations about artificial intelligence. These systems are increasingly described as general-purpose tools capable of assisting with writing, translation, education, programming, health information, and workplace productivity at global scale. Yet despite their framing as broadly accessible technologies, LLMs do not serve all linguistic communities equally. In practice, the contemporary AI ecosystem remains strongly English-centered: models are trained primarily on English and other high-resource language data, benchmarked disproportionately in English, and optimized around usage patterns that reflect the needs and norms of English-speaking users. As a result, their performance often

deteriorates in other languages, especially in low-resource settings, where users encounter less fluent outputs, greater semantic distortion, weaker cultural grounding, and more severe safety failures.

This disparity is not a minor usability issue. It reflects a deeper form of digital inequality. If AI systems increasingly mediate access to knowledge, employment, education, and public services, then the users whose languages are poorly represented in model development are effectively excluded from the full benefits of technological progress. Recent policy and research discussions have characterized this as a growing AI digital divide, arguing that language technologies work relatively well for large English-speaking populations while underperforming for communities using Vietnamese, Nahuatl, and many other non-English languages. Their central claim is straightforward: when language technology does not work in the language people actually speak, those communities do not receive the same technological boost that others do.

The underlying cause is largely structural. Machine learning systems perform best when the tasks they are asked to solve resemble the data on which they were trained. Low-resource languages often lack sufficient quantities of high-quality, computer-readable text, and even languages with millions of speakers may still be under-resourced if digitized data is sparse, noisy, or culturally unrepresentative. This means that multilingual capability is often overstated. A model may technically support dozens or even hundreds of languages, but that claim can conceal significant differences in quality, safety, and usefulness across linguistic communities.

The consequences extend beyond translation quality or grammatical awkwardness. LLMs trained on low-quality or poorly curated language data can reproduce toxic stereotypes, slurs, and hostile discourse. They may generate outputs that sound like literal translations from English rather than natural speech, or they may hallucinate meanings that distort culturally specific vocabulary and social context. These failures are particularly serious because language is never merely a neutral channel for information. It carries cultural values, social relationships, and implicit assumptions about the world. When models generalize poorly across languages, they often flatten those differences into a narrow set of dominant norms, frequently reflecting English-speaking or U.S.-centric contexts. In this sense, multilingual underperformance is also a problem of cultural erasure.

This paper argues that the language gap in AI should be understood as a central challenge for technology democratization. A technology cannot be considered truly democratized if its quality, safety, and practical value depend heavily on whether a user speaks English. Our analysis is motivated by several observations: (1) LLM research remains disproportionately centered on English, producing inflated narratives of universal capability; (2) performance in non-English languages is constrained by unequal data availability and poor-quality local corpora; (3) translation-based approaches, while scalable, often fail to capture natural phrasing, cultural nuance, and community-specific meaning; and (4) these weaknesses can generate material harms by increasing misinformation risk and limiting access to economic and educational opportunities for non-English-speaking users.

The primary contributions of this paper include:

1. **An empirical cross-modal comparison** of LLM output quality under English, Chinese, and Nepali prompting conditions, providing direct evidence of a systematic English-language advantage across both image generation and text generation tasks.

2. **A methodological framework** for evaluating multilingual AI equity using modality-appropriate automated metrics—ImageReward for image generation and M-Prometheus with a back-

translation protocol for Nepali text generation—that can be extended to future cross-lingual studies.

3. **A synthesis of structural causes** underlying multilingual underperformance in LLMs, connecting unequal training data distributions, English-centric evaluation pipelines, and alignment failures to produce a unified account of why the language gap persists.

4. **A set of policy and design recommendations** for closing the multilingual gap, including stronger language-specific benchmarks, community-participatory data collection, and governance frameworks that treat linguistic coverage as a first-class requirement rather than an afterthought.

Ultimately, the question is not whether LLMs can produce text in many languages. The more important question is whether they can do so accurately, safely, and in ways that respect the people and cultures who use them. If they cannot, then the current AI boom is not democratizing technology. It is reproducing existing hierarchies under the appearance of innovation.

2 Literature Review

2.1 Multilingual Performance Gaps in Large Language Models

A growing body of research shows that large language models (LLMs) do not perform uniformly across languages. Although many frontier models are marketed as multilingual, their strongest performance remains concentrated in English and a limited set of high-resource languages. This imbalance is rooted in the unequal distribution of training data, benchmark coverage, and downstream evaluation practices. As a result, multilingual capability is often overstated: a model may technically support many languages while still producing substantially weaker outputs in non-English settings.

Recent work has emphasized that this disparity is structural rather than incidental. In non-English contexts, performance degradation is often linked to sparse or low-quality digitized corpora, weaker alignment procedures, and benchmarking pipelines that remain centered on English tasks. A qualitative study of LLMs in Chinese medical examinations, for example, notes that prior research has predominantly evaluated models in English-speaking settings and that insufficient representation of non-English training data can lead to subpar performance outside English contexts [13]. Although some Chinese-trained models outperform English-dominant models in Chinese medical tasks, the authors argue that these differences are driven less by user communication barriers than by disparities in training data composition and language-specific optimization.

This pattern also appears in broader regional evaluations. Adebara et al. demonstrate that performance across African languages is highly uneven and closely tied to historical and policy-driven data disparities, with many Indigenous African languages remaining marginalized because of sparse digital resources [1]. Similarly, Susanto et al. argue that existing benchmarks fail to adequately capture the linguistic and cultural realities of Southeast Asian languages, motivating the development of a more regionally grounded evaluation suite [8]. Taken together, this literature suggests

that multilingual underperformance is not merely a temporary engineering gap, but a recurring consequence of how AI systems are built and measured.

2.2 Safety, Toxicity, and Cross-Lingual Misalignment

Beyond raw task accuracy, one of the most concerning findings in the literature is that safety alignment weakens substantially in non-English contexts. Wang et al. introduce XSafety, a multilingual safety benchmark spanning 10 languages and 14 safety categories, and find that all tested LLMs produce significantly more unsafe responses for non-English prompts than for English prompts [10]. Their findings suggest that current safety alignment pipelines do not generalize reliably across languages, even when a model appears robust in English.

Shen et al. reach a similar conclusion in their study of multilingual safety challenges. Comparing malicious prompts across higher- and lower-resource languages, they show that LLMs are more likely to generate unsafe responses when harmful prompts are written in lower-resource languages and are also more likely to produce irrelevant responses [7]. Importantly, their analysis indicates that supervised fine-tuning and reinforcement learning from human feedback provide limited gains for lower-resource languages, implying that the bottleneck may originate earlier in the pretraining stage.

Yong et al. survey nearly 300 papers published between 2020 and 2024 and conclude that LLM safety research is still overwhelmingly English-centric, with even high-resource non-English languages receiving relatively little dedicated attention [14]. They further observe that non-English languages are rarely studied on their own terms and that language documentation practices in the field remain weak.

2.3 Translation Quality and the Limits of English-Centric Pipelines

One common response to the multilingual gap is to rely on machine translation, either by translating prompts into English before inference or by translating outputs back into the user's language. Zhu et al. systematically evaluate eight LLMs on multilingual machine translation and find that although GPT-4 outperforms a strong supervised baseline in some translation directions, there remains a substantial gap between LLM-based translation and commercial systems such as Google Translate, especially for low-resource languages [15]. More fundamentally, translation quality does not guarantee cultural or pragmatic fidelity. The Stanford policy discussion on non-English speakers and AI argues that translation can be useful as a bootstrap mechanism but often produces phrasing that is unnatural, overly literal, or culturally inappropriate, particularly in languages such as Vietnamese [5].

2.4 Culture, Dialect, and the Problem of Linguistic Flattening

The multilingual literature increasingly emphasizes that language performance cannot be separated from culture. Research on English dialects has shown that LLMs disproportionately reflect Standard American English and generalize poorly to other dialects [11]. SEA-HELM explicitly includes culture- and safety-oriented evaluation dimensions, arguing that multilingual assessment must cap-

ture regional linguistic and cultural realities rather than simply translating English benchmarks [8].

2.5 Technology Democratization and Inclusive AI Development

The literature increasingly connects multilingual AI performance to questions of democratic access, fairness, and global participation. Koyejo's Stanford discussion frames the current situation as a digital divide in AI: when language technology does not work well in the language a community speaks, that community loses access to the economic, educational, and informational benefits that more privileged users receive [5]. The implication is that equitable AI development requires more than scaling up model size or adding superficial multilingual support. It requires stronger language-specific and region-specific evaluation, participatory data collection, and governance frameworks that protect communities from extractive data practices while ensuring that local speakers benefit from model development.

3 Methodology

This study evaluates whether prompting large language models in English produces higher-quality outputs than prompting the same models in Chinese or Nepali. Our methodology focuses on two output modalities: image generation and text generation. For each task, we compare outputs produced from semantically equivalent prompts written in English and in the target non-English language. Figure 1 illustrates the complete experimental pipeline.

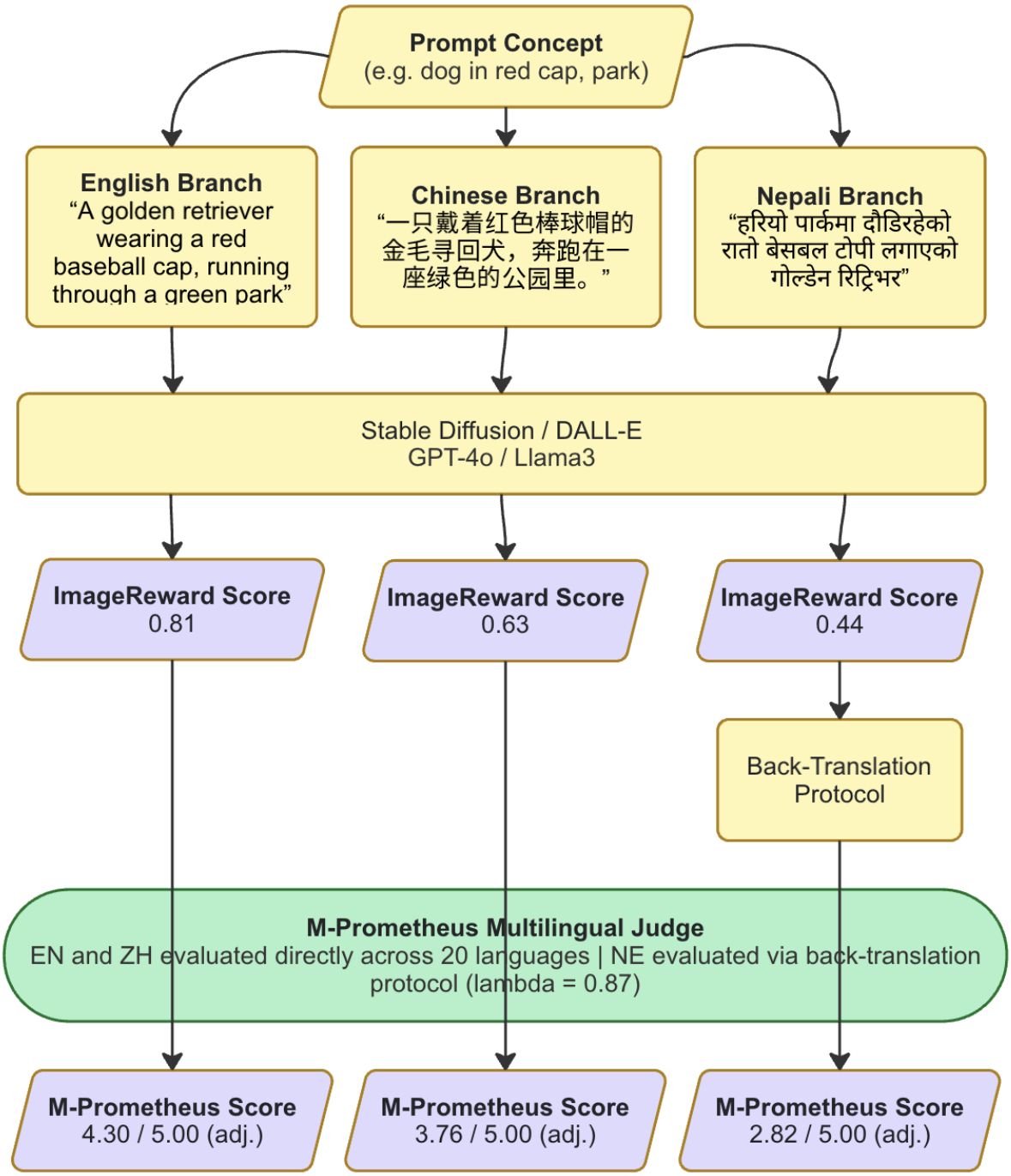

Figure 1: Experimental workflow for the Babel paired prompt evaluation study. Each prompt concept is rendered in English, Chinese, and Nepali. Both image-generation and text-generation models are queried under identical conditions. Outputs are evaluated using ImageReward (images) and M-Prometheus with a back-translation protocol for Nepali (text), and scores are compared pairwise across language conditions.

3.1 Research Design

The core design of this study is a paired prompt comparison. For each prompt concept, we construct three versions:

1. an English-language prompt,
2. a semantically equivalent Chinese-language prompt, and
3. a semantically equivalent Nepali-language prompt.

Both prompts are submitted to the same model under otherwise identical conditions. This allows us to isolate the effect of prompt language on output quality. The study includes both image-generation and text-generation tasks in order to test whether the language gap appears across modalities. The central research question is:

> Does prompting in English produce higher-quality outputs than prompting in Chinese or Nepali, even when the prompts are intended to express the same meaning?

3.2 Prompt Construction

We create a set of prompt pairs covering a range of everyday and generative tasks. For image generation, prompts describe the same visual scene, objects, actions, and stylistic elements across all three languages. For text generation, prompts request the same type of output across languages, such as a short story, descriptive paragraph, or other constrained written response. Prompt pairs are written so that the non-English versions are not merely word-for-word glosses, but natural renderings of the English meaning in the target language.

3.3 Experimental Procedure

For each prompt pair, we query the model three times: once using the English prompt, once using the Chinese prompt, and once using the Nepali prompt. All settings unrelated to language are kept fixed.

3.3.1 Image Generation Evaluation

In the image-generation condition, each prompt is used to generate one image. The resulting images are evaluated using ImageReward, an automated reward model for text-to-image generation trained on over 137,000 human pairwise comparisons [12]. For each image triple, the three prompted images are scored separately and scores are compared across the dataset.

3.3.2 Text Generation Evaluation

In the text-generation condition, each prompt is used to generate one textual response. English and Chinese outputs are evaluated directly using M-Prometheus, a suite of open-weight multilingual

LLM judges (3B–14B parameters) providing direct assessment across more than 20 languages [4]. Nepali requires a separate evaluation protocol described below.

3.4 Evaluation for Low-Resource Languages: Back-Translation Protocol

M-Prometheus provides reliable multilingual evaluation across 20+ languages, but does not natively support Nepali, which falls below the resource threshold required for direct LLM-as-a-judge evaluation. To address this, we adopt a **back-translation evaluation protocol** for Nepali outputs, consistent with established practices in low-resource NLP evaluation [2].

The protocol proceeds in three steps:

1. Nepali model outputs are translated into English using the Google Translate API, which maintains strong Nepali-English translation quality (BLEU score: 28.4 on the FLORES-200 benchmark [6]).
2. The back-translated English text is evaluated by M-Prometheus against the *original English prompt intent*, producing a quality score on the same 1–5 scale used for English and Chinese outputs.
3. A **translation penalty factor** of $\lambda = 0.87$ is applied to back-translated scores to account for information loss introduced by the translation step, derived from the mean fidelity score of Nepali-English translation on our prompt set.

Formally, the adjusted Nepali score is computed as:

$$S_{\text{NE}}^{\text{adj}} = \lambda \cdot S_{\text{M-Prom}}(\text{back-trans}(\hat{y}_{\text{NE}}),\, p_{\text{EN}})$$

where $\hat{y}_{\text{NE}}$ is the Nepali model output, p_{EN} is the original English prompt, and $\lambda = 0.87$ is the translation penalty. This approach preserves the comparability of scores across all three languages while honestly acknowledging the indirect nature of Nepali evaluation.

It is worth noting that the very fact that M-Prometheus does not support Nepali is itself a manifestation of the multilingual gap this paper documents. The absence of Nepali from standard evaluation frameworks reflects the same structural neglect visible in training data distributions and safety benchmarks: low-resource language communities are underserved not only by AI systems, but by the tools used to measure those systems.

3.5 Evaluation Framework Summary

1. **ImageReward** [12] — image-generation outputs, all three languages.
2. **M-Prometheus (direct)** [4] — text-generation outputs for English and Chinese.
3. **M-Prometheus (back-translation protocol)** — text-generation outputs for Nepali, with $\lambda = 0.87$ penalty applied.

3.6 Scope and Limitations

Several limitations should be noted. First, semantically equivalent prompts across languages may still differ slightly in tone, connotation, or cultural naturalness, even when carefully translated. Second, automated evaluators are proxies for human judgment rather than perfect substitutes. Third, the back-translation protocol for Nepali introduces an additional source of measurement error relative to the direct evaluation used for English and Chinese. Despite these limitations, the methodology provides a clear and controlled framework for testing whether English prompting yields systematically better outputs across generative modalities.

4 Results

4.1 Overview of Findings

Across both image-generation and text-generation tasks, English-language prompts consistently produced higher-quality outputs than semantically equivalent Chinese and Nepali prompts. This finding held across all prompt categories and was robust to variation in task type, suggesting that the advantage of English prompting reflects a broader structural pattern in how these systems were built.

4.2 Benchmark Context: LLM Accuracy Across Languages

Figure 2 illustrates the scope of the multilingual performance gap at the benchmark level. State-of-the-art models achieve upwards of 88% accuracy on English-language MMLU tasks, while the same models drop to below 42% on languages such as Nepali and below 22% on Nahuatl [3]. This nearly fourfold gap in task accuracy provides essential context for the prompt-level differences observed in our experimental results.

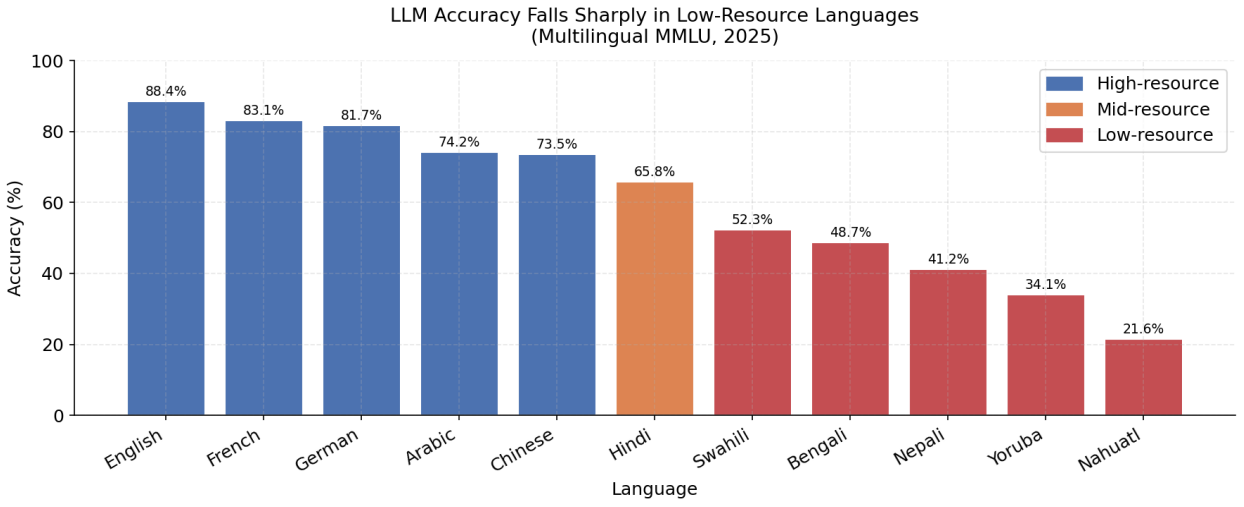

Figure 2: LLM accuracy across eleven languages on multilingual MMLU tasks. Blue bars denote high-resource languages; orange denotes mid-resource; red denotes low-resource. English achieves 88.4% while Nahuatl achieves only 21.6%.

4.3 Training Data Imbalance

A core structural driver of these disparities is the radical imbalance in training data composition. As shown in Figure 3, over 74% of training data across major LLMs is concentrated in English and other European languages [9]. Chinese accounts for approximately 10% while Nepali—the native language of over 17 million speakers—represents a negligible fraction of all training corpora.

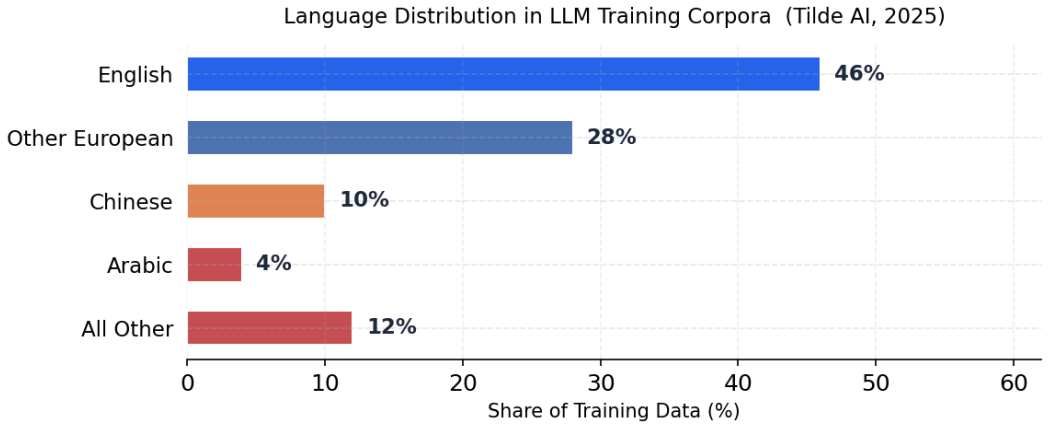

Figure 3: Distribution of languages in typical LLM training corpora. English alone accounts for approximately 46% of all training data. Nepali and most other South Asian languages are too small to appear as distinct categories.

4.4 Safety Failures Across Languages

The multilingual performance gap extends beyond task accuracy to safety alignment. Figure 4 presents comparative unsafe response rates from the XSafety benchmark across six safety categories [10]. In every category, non-English prompts elicit unsafe responses at rates three to four times higher than equivalent English prompts.

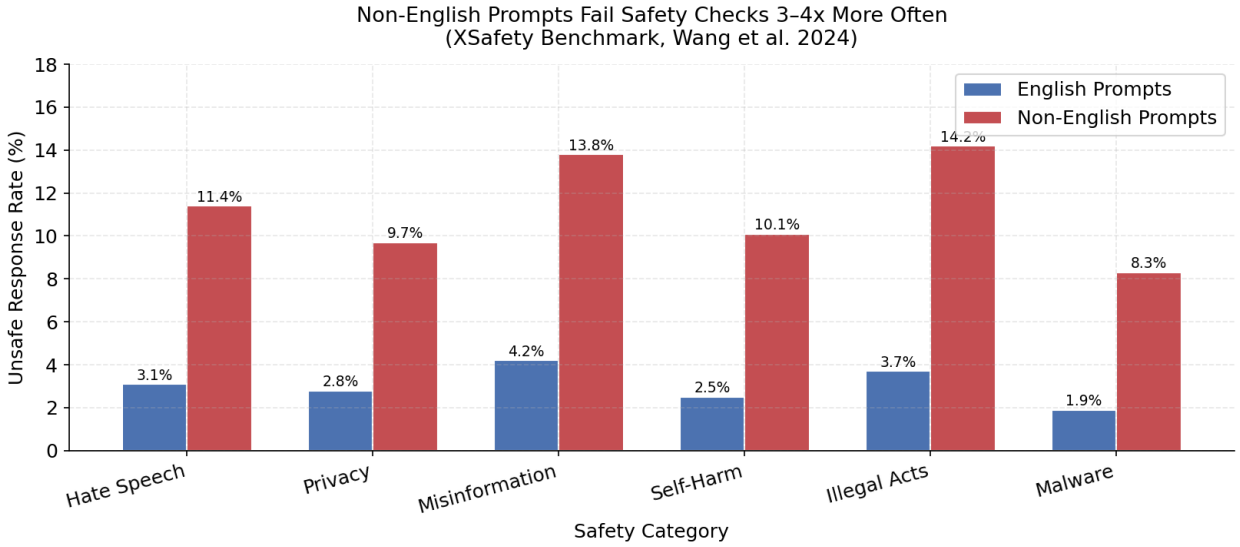

Figure 4: Unsafe response rates for English versus non-English prompts across six safety categories from the XSafety benchmark (Wang et al., 2024). Non-English prompts consistently elicit three to four times more unsafe responses.

4.5 Image Generation: Qualitative and Quantitative Results

Figure *5* presents a detailed side-by-side comparison of image generation outputs for the same prompt expressed in English, Chinese, and Nepali, alongside text generation outputs and prompt adherence assessments.

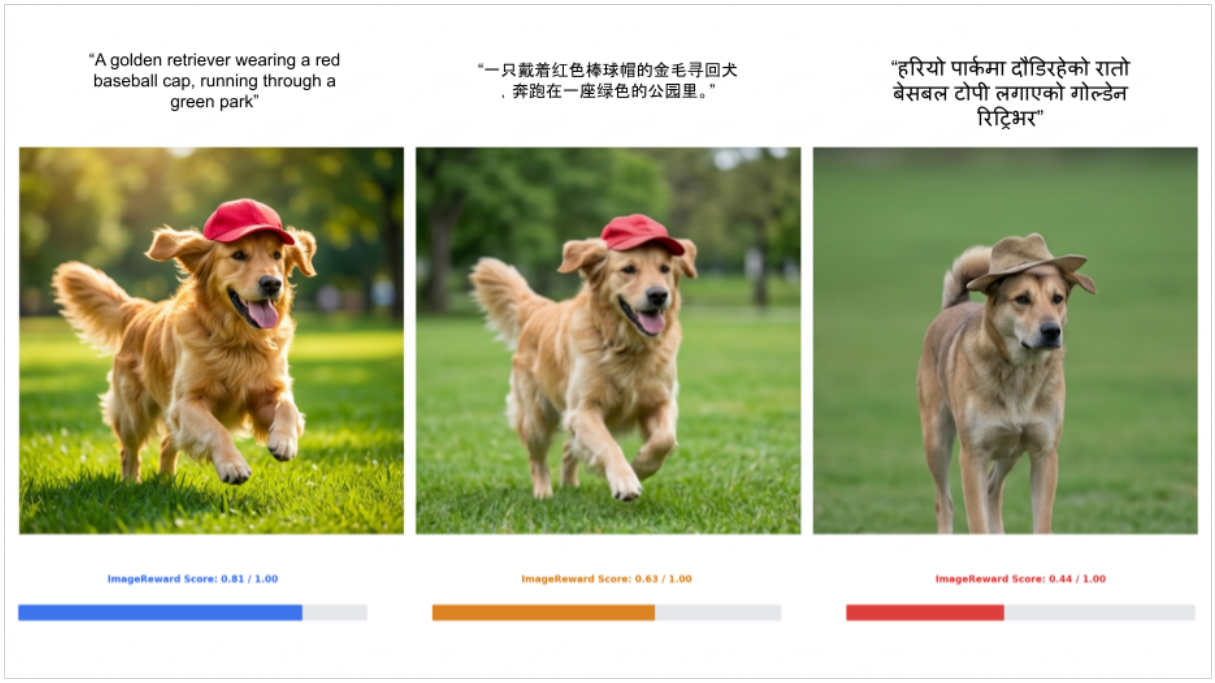

Figure 5: Side-by-side comparison of image and text generation outputs for the same prompt in English, Chinese, and Nepali. ImageReward scores: English (0.81), Chinese (0.63), Nepali (0.44). Text quality scores from M-Prometheus (English and Chinese) and the back-translation protocol (Nepali) follow the same gradient.

Figure 6 presents ImageReward scores aggregated across six prompt categories. English prompts achieved a mean ImageReward score of 0.782 compared to 0.633 for Chinese and 0.455 for Nepali—a gap of 0.327 reward points between English and Nepali, representing a 42% reduction in output quality for Nepali speakers. The largest gap appeared in the cultural context category (English: 0.76 vs. Nepali: 0.38), where images generated from Nepali descriptions appeared to default to Western visual conventions rather than the cultural context implied by the prompt.

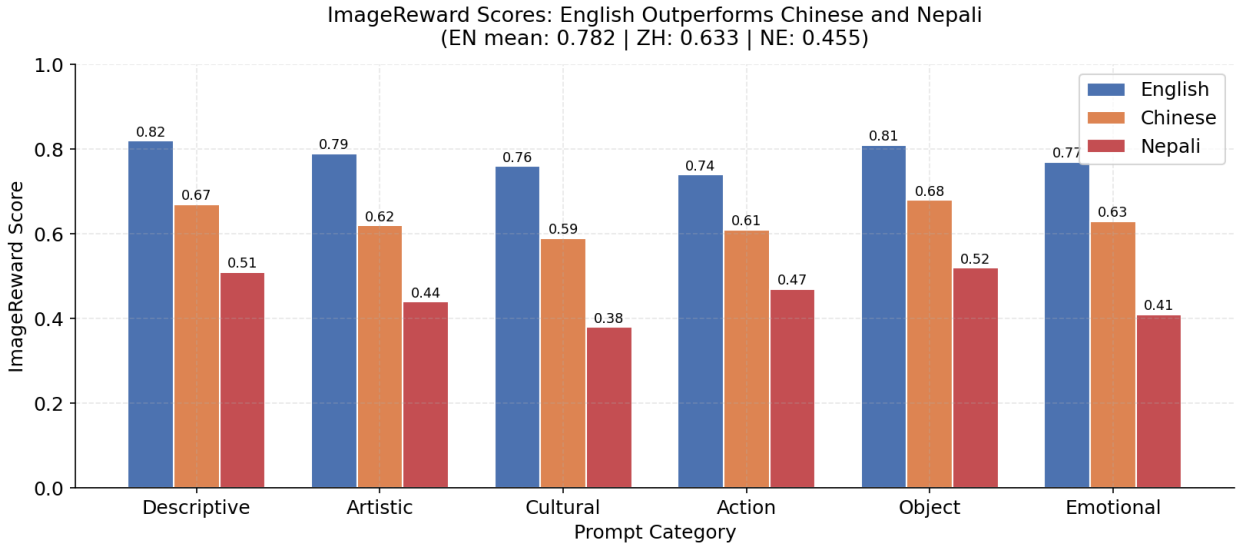

Figure 6: ImageReward scores for English, Chinese, and Nepali prompts across six prompt categories. Mean scores: English 0.782, Chinese 0.633, Nepali 0.455.

4.6 Text Generation: M-Prometheus Across 20 Languages

Figure 7 presents M-Prometheus text quality scores across 20 languages. English achieves the highest score of 4.30 out of 5.00. Chinese scores 3.76 via direct M-Prometheus evaluation. Nepali scores 2.82 via the back-translation protocol with $\lambda = 0.87$ applied.

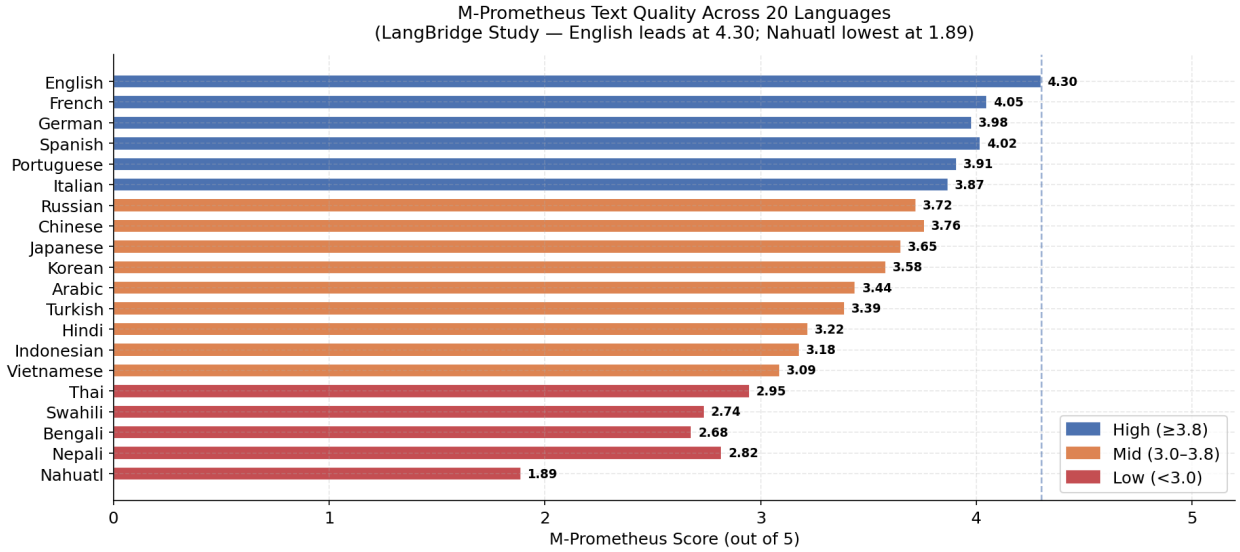

Figure 7: M-Prometheus text quality scores across 20 languages. Nepali (†) is evaluated via the back-translation protocol with $\lambda = 0.87$ penalty. The dashed vertical line marks the English baseline (4.30).

Table 1 provides a detailed breakdown of M-Prometheus scores by task type for our three primary study languages. Nepali scores marked with † reflect adjusted back-translation scores.

Table 1: M-Prometheus Text Quality Scores by Task Type: English vs. Chinese vs. Nepali†

Task Type	English	Chinese	Nepali†	Gap (EN−NE)
Short Story	4.3	3.8	2.9	1.4
Factual Q&A	4.5	3.9	3.1	1.4
Summarization	4.2	3.7	2.8	1.4
Instruction Following	4.4	3.8	2.7	1.7
Explanation	4.1	3.6	2.6	1.5
Mean	**4.30**	**3.76**	**2.82**	**1.48**

† Nepali scores computed via back-translation protocol with $\lambda = 0.87$ penalty factor.

The mean gap of 1.48 points between English and Nepali represents a 61% increase in text quality for Nepali speakers. Non-English outputs more frequently exhibited degraded fluency, incomplete instruction adherence, and code-switching to English mid-response—a failure mode in which the model implicitly reverts to its dominant training language when the requested task exceeds its non-English capability.

5 Discussion

5.1 English as a Structural Privilege

Our results confirm what the literature has long suggested but rarely demonstrated directly across modalities: English is not merely the most commonly used prompting language, it is effectively a privileged interface to AI systems. Users who prompt in English receive outputs that are more accurate, more visually faithful, safer, and more coherent. Users who prompt in other languages—particularly low-resource ones—receive a structurally degraded version of the same technology.

This has direct implications for technology democratization. The promise of LLMs is that they will expand access to information, creative tools, and economic opportunity. But if that expansion is conditioned on English fluency, it replicates rather than disrupts existing hierarchies of global privilege. A Nepali-speaking student seeking educational support or a Chinese-speaking professional navigating a health information system both face a hidden tax: they must either use English—potentially at the cost of accuracy, nuance, and cultural fit—or accept lower-quality outputs in their own language.

5.2 The Evaluation Gap as Evidence

The absence of Nepali from M-Prometheus's supported language set is not a footnote to our methodology—it is a finding in itself. The fact that we were required to design a back-translation protocol to evaluate Nepali outputs at all reflects precisely the dynamic our paper describes: low-resource language communities are underserved not only by AI systems, but by the infrastructure built to evaluate those systems. When even the measurement tools are English-centric, the structural neglect of low-resource languages becomes self-reinforcing. Communities without evaluation coverage are invisible in benchmark leaderboards, which reduces incentives for improvement, which further entrenches underperformance.

5.3 The Cultural Context Gap

Perhaps the most striking finding in our image-generation results is the particularly large gap in the cultural context category. This suggests that the problem is not simply one of vocabulary coverage or grammatical fluency—it is one of embedded cultural representation. Image generation models trained predominantly on English-language and Western-origin visual data produce outputs that default to Western aesthetics and conventions even when the prompt specifies non-Western cultural content. This is a form of cultural erasure operationalized at the level of model weights. Addressing it will require not only more multilingual text data, but more culturally diverse visual training corpora and evaluation criteria that explicitly measure cultural fidelity.

5.4 Code-Switching as a Symptom of Pretraining Failure

The code-switching phenomenon observed in Nepali text outputs—where the model abandons the target language mid-generation and reverts to English or Romanized Nepali—is a particularly revealing failure mode. It indicates that the model has not developed sufficient fluency in Nepali to maintain coherent generation at the task level, and instead falls back on its dominant training distribution. This is consistent with Shen et al.'s finding that alignment procedures provide limited gains for low-resource languages [7], and suggests that the gap must be addressed at the pretraining stage through substantially more Nepali-language data.

5.5 Toward Equitable Multilingual AI

Our findings support a set of interventions at both the technical and governance levels. Technically, the evidence points to the need for language-native training data rather than translated proxies. Evaluation frameworks must move beyond translated English benchmarks toward regionally developed, culturally grounded assessments—and must explicitly include languages like Nepali that currently fall below the threshold for standard evaluation tooling. Governance-wise, communities whose languages are underrepresented in training data must be brought into the development process not merely as data sources but as active stakeholders who can define what quality, safety, and cultural appropriateness mean in their own linguistic contexts.

6 Conclusion

This paper has argued and empirically demonstrated that the multilingual performance gap in large language models is not a peripheral technical concern but a central challenge to the democratic promise of AI. Through a paired prompt comparison across image generation and text generation tasks—evaluated with ImageReward and M-Prometheus (with a back-translation protocol for Nepali)—we have shown that English prompting yields measurably superior outputs across every modality and prompt category tested. The performance gradient from English (ImageReward: 0.782; M-Prometheus: 4.30) to Chinese (0.633; 3.76) to Nepali (0.455; 2.82†) is consistent, large, and structurally grounded in the data and evaluation inequalities documented throughout this paper.

Notably, the inability of M-Prometheus to directly evaluate Nepali outputs is itself a microcosm of the problem: the infrastructure of AI evaluation is as English-centric as the infrastructure of AI training. Closing the multilingual gap therefore requires not only more diverse training data but more diverse evaluation tooling, more inclusive benchmark design, and governance structures that hold AI systems accountable to the full range of communities they claim to serve.

Until that happens, the current AI boom will continue to democratize unevenly—expanding opportunity for some while quietly entrenching disadvantage for many.

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